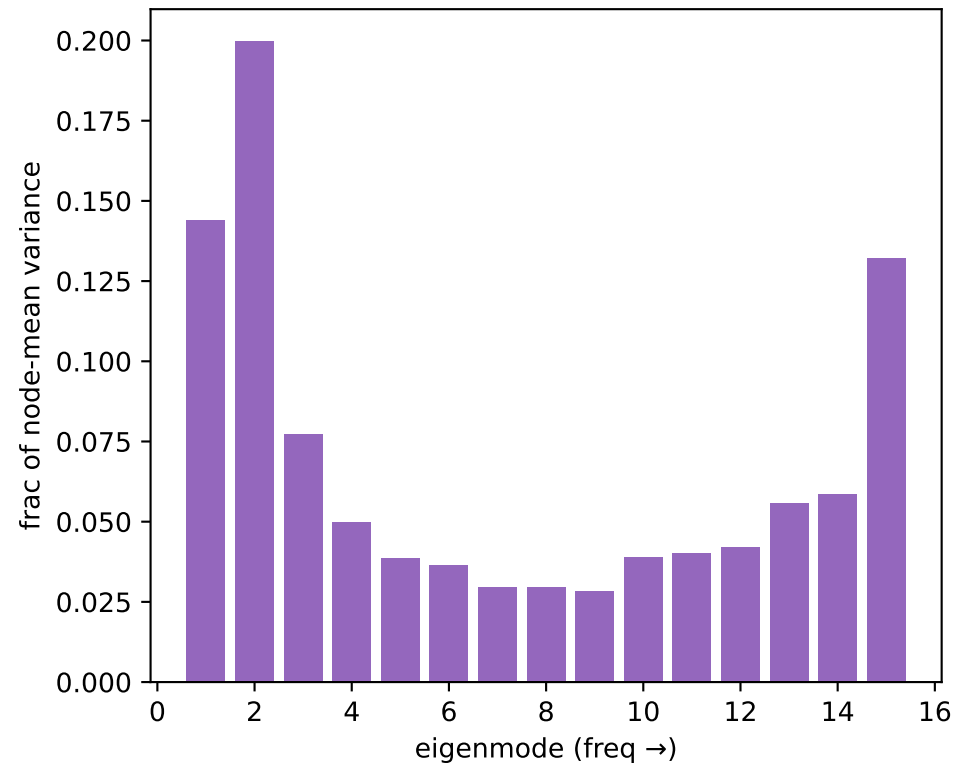
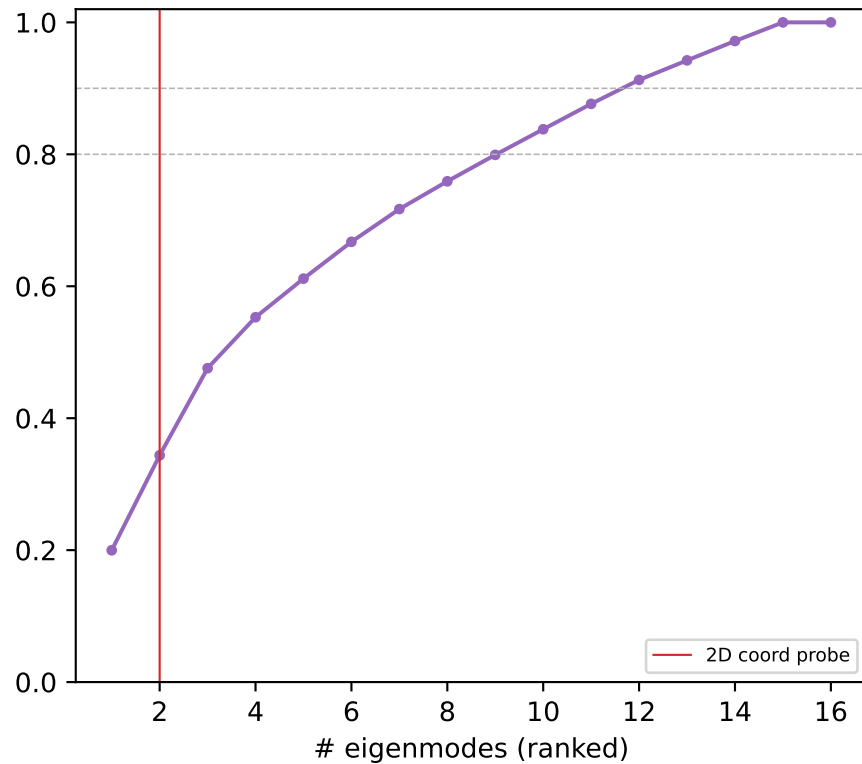


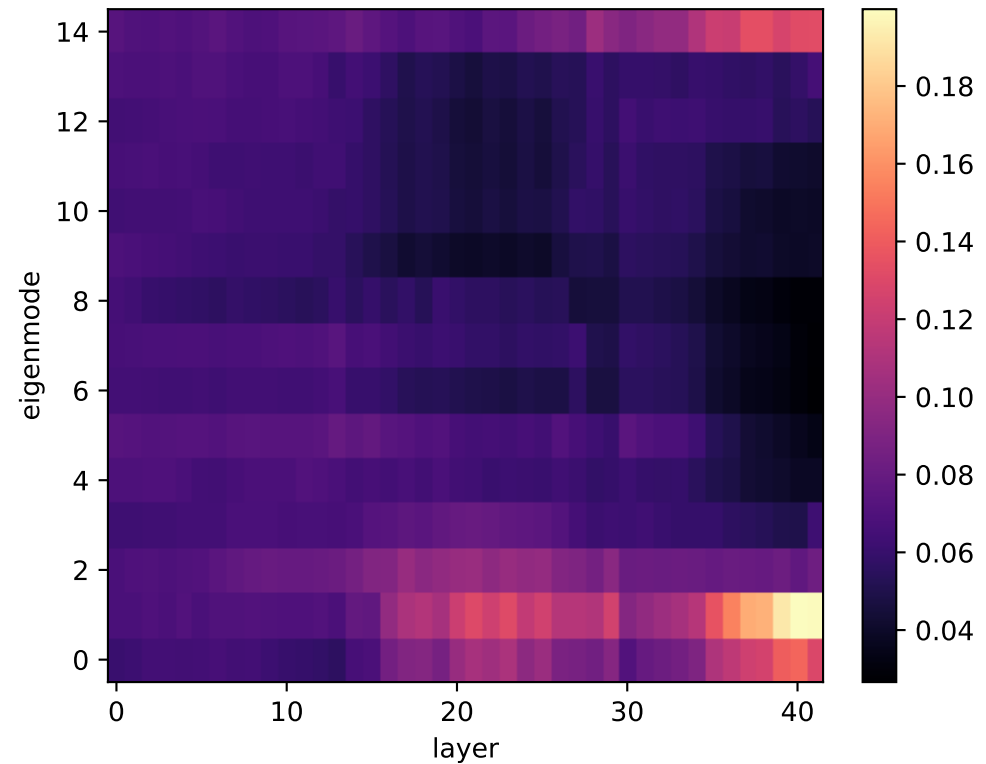
Gemma square_grid L40: power per Laplacian mode



cumulative variance vs #modes

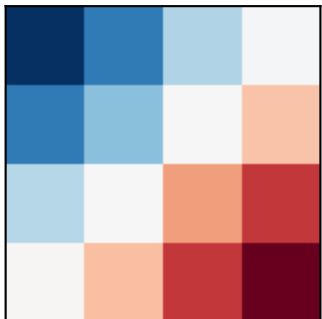


power: eigenmode × layer

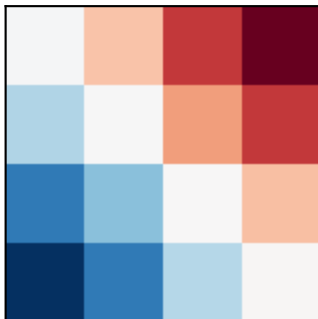


Gemma square_grid: Laplacian eigenmode DIVIDERS (% = variance share at L40)

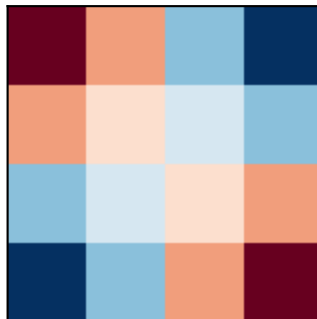
mode 1 $\lambda=0.59$ (14%)



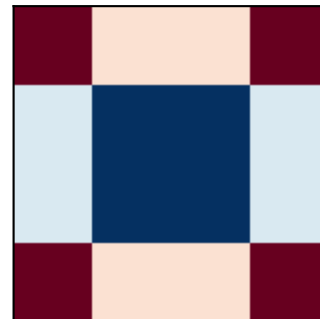
mode 2 $\lambda=0.59$ (20%)



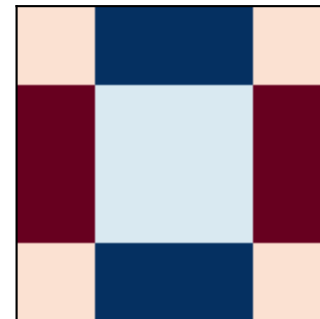
mode 3 $\lambda=1.17$ (8%)



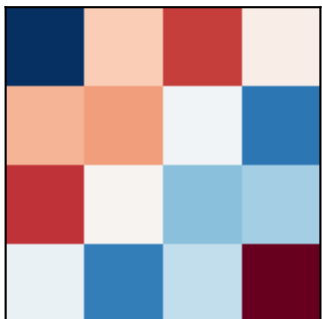
mode 4 $\lambda=2.00$ (5%)



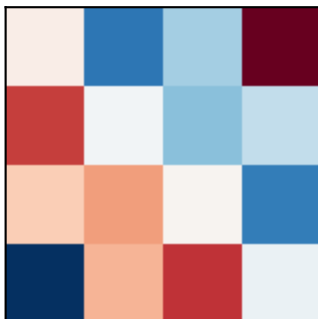
mode 5 $\lambda=2.00$ (4%)



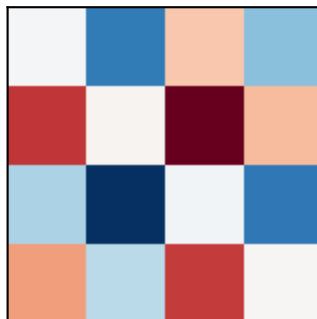
mode 6 $\lambda=2.59$ (4%)



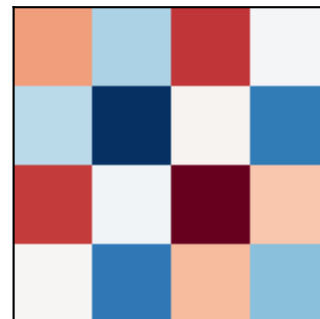
mode 7 $\lambda=2.59$ (3%)



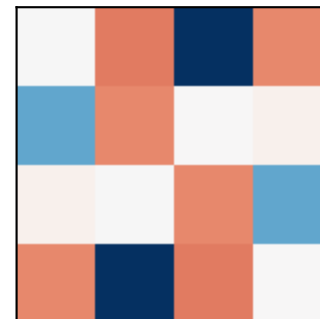
mode 8 $\lambda=3.41$ (3%)



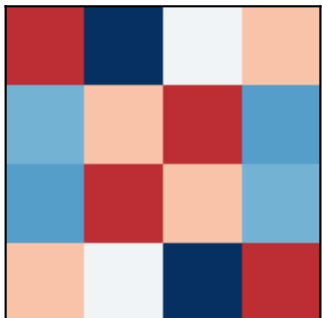
mode 9 $\lambda=3.41$ (3%)



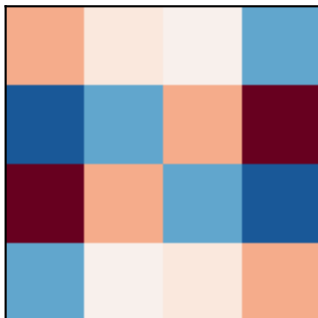
mode 10 $\lambda=4.00$ (4%)



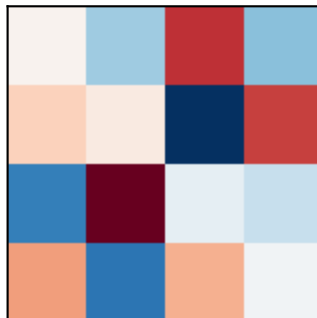
mode 11 $\lambda=4.00$ (4%)



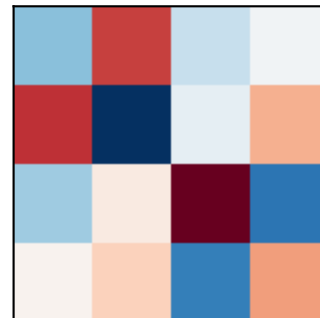
mode 12 $\lambda=4.00$ (4%)



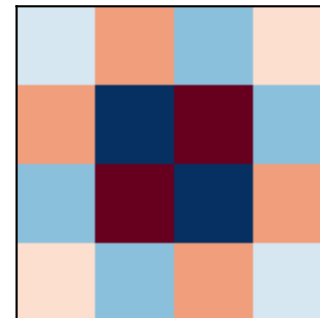
mode 13 $\lambda=5.41$ (6%)



mode 14 $\lambda=5.41$ (6%)

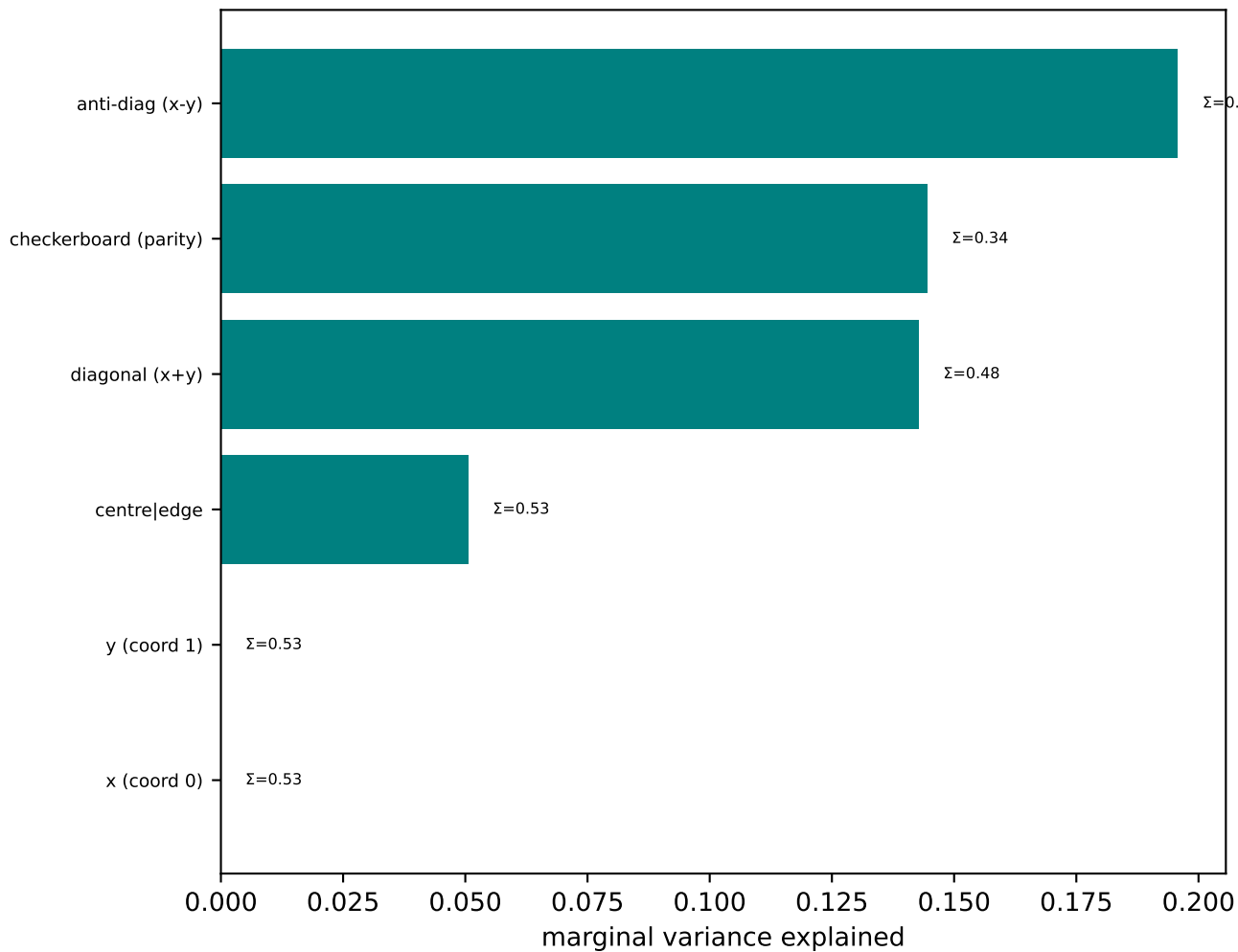


mode 15 $\lambda=6.83$ (13%)

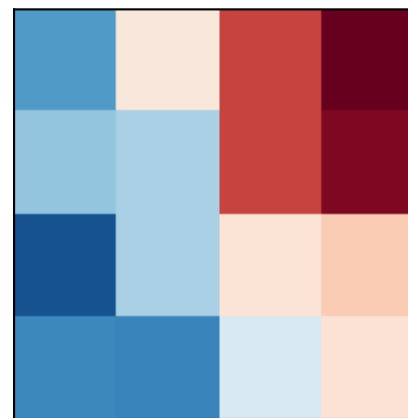


Gemma square_grid: named-cut basis (left) & model's own top axes (right)

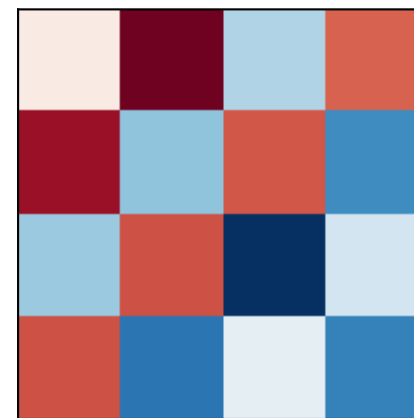
named cuts, greedy orthogonal (cumulative Σ)



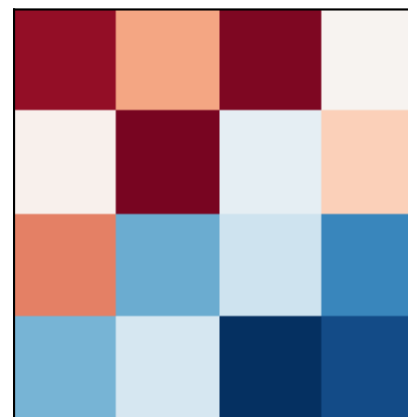
PC0 (22%)
 $\approx$ anti-diag (x-y)



PC1 (16%)
 $\approx$ checkerboard (parity)



PC2 (14%)
 $\approx$ diagonal (x+y)



PC3 (8%)
 $\approx$ centre|edge

